

The Impact of Socioeconomic Factors on DNR Orders Adjusting for Clinical Prognosis

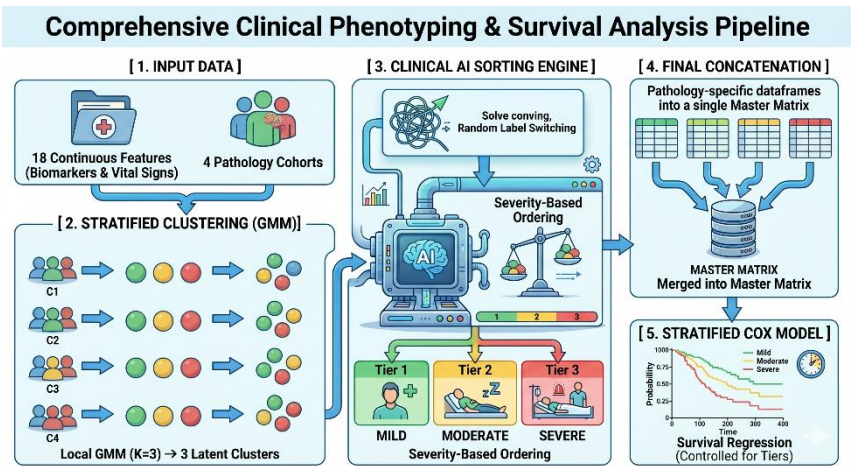
M. Albergoni, A. Bekaj, R. R. Rengifo Ybarra, T. Spadari, M. Torcasio

Introduction

DNR orders are critical in the management of critically ill patients. Ideally, these decisions should be strictly dictated by clinical prognosis, however growing evidence suggests that non-clinical elements, can heavily bias these decisions.

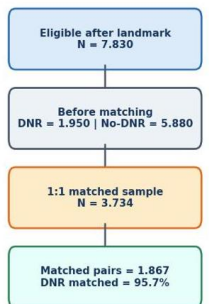
Methods

- Temporal Auditing & Landmark Analysis:** Implemented at Day 3 to eliminate immortal time bias and resolve chronological data-entry paradoxes, securing Cox model convergence → 7830.
- Hierarchical clinical cleaning strategy before imputation:**
 - Pure instrumental zeros**, such as albumin, blood urea nitrogen, creatinine, white blood cell count and temperature → biologically implausible, converted to NaN
 - Terminal vital zeros**, such as mean blood pressure, heart rate, respiratory rate and PaO₂/FiO₂ ratio, were evaluated through a **clinical coherence check**.
 - Non-zero** values were screened using predefined biological plausibility limits:
 - Outside absolute physiological limits → NaN
 - Within soft/statistical outlier ranges → **retained** or **winsorized**
- Clinical coherence** was assessed using the patient's **diagnostic context**, including ARF, cirrhosis, CHF, COPD and coma, together with neurological severity measured by the SUPPORT Coma Score.
- Remaining missing values were imputed using a **non-linear MICE** approach based on **IterativeImputer** with **ExtraTreesRegressor**.



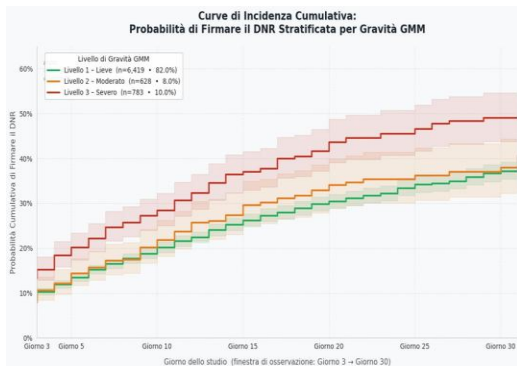
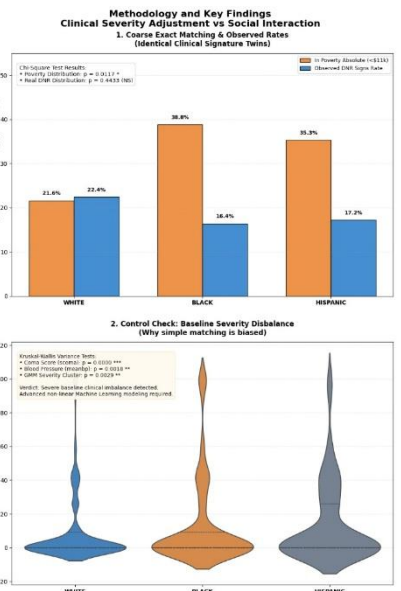
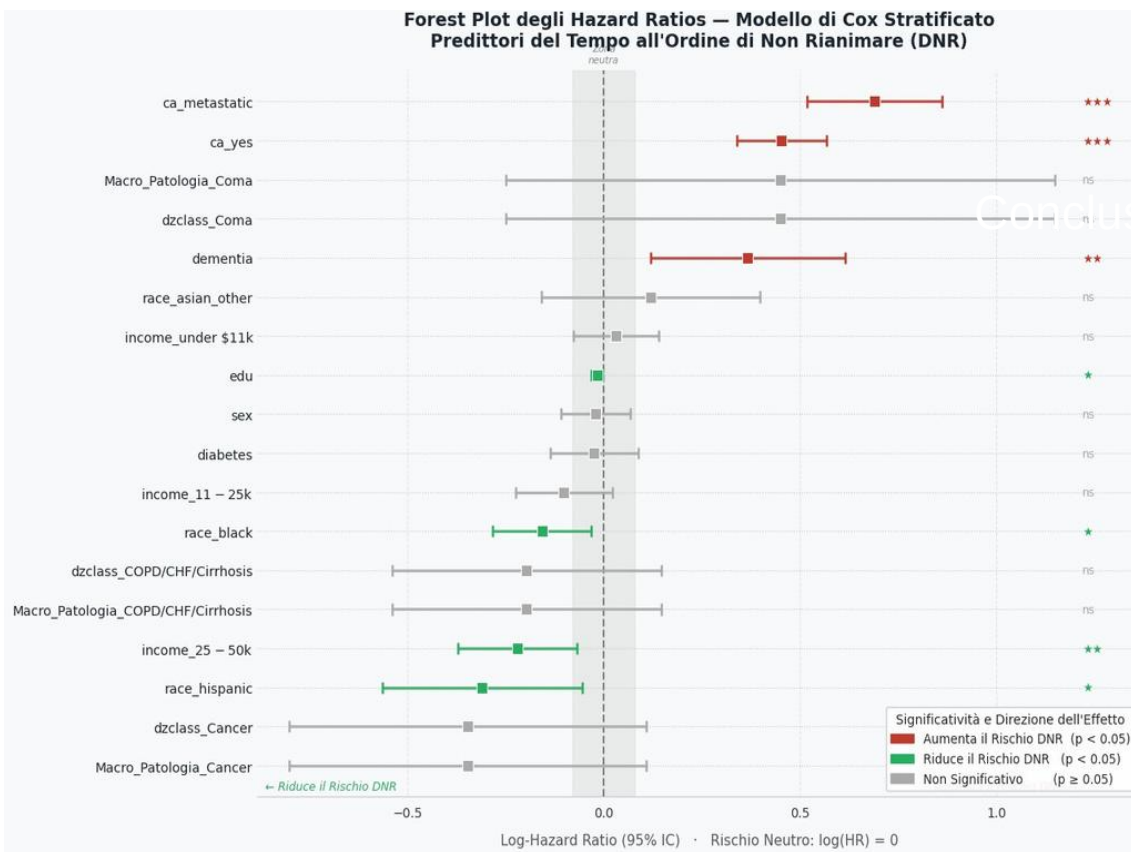
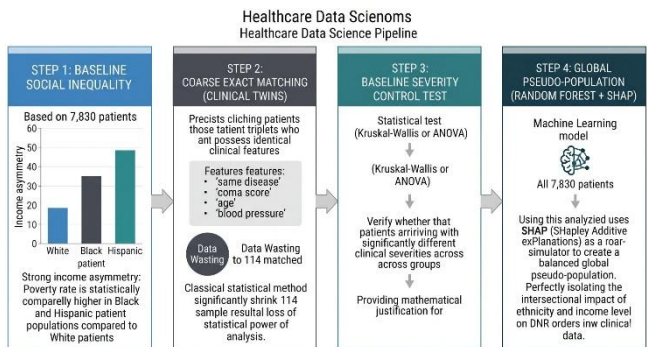
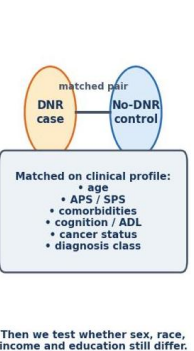
Matching strategy and regression results: DNR decisions after clinical balancing

A. Cohort construction



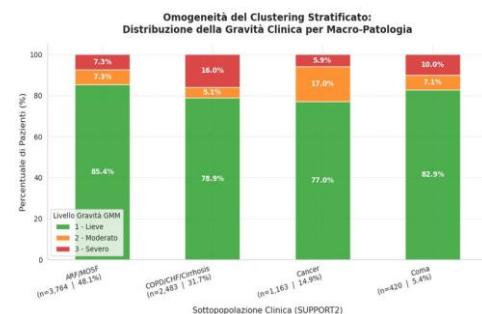
Goal: compare DNR and No-DNR patients with similar clinical severity.

B. What matching does



High Confidence: 90%+ classification certainty achieved for 92.64% of ARF/MOSF patients.

ORDINAMENTO CLINICO DEI FENOTIPI (CLUSTER)			
LEGENDA: 1=Lieve, 2=Moderato, 3=Severo			
MACRO-PATOLOGIA	LIVELLO 1 (LIEVE)	LIVELLO 2 (MODERATO)	LIVELLO 3 (SEVERO)
ARF/MOSF	Raw C0 Score: 0.23	Raw C1 Score: 4.05	Raw C2 Score: 5.07
COPD/CHF/CIRRHOSIS	Raw C2 Score: 0.89	Raw C1 Score: 4.51	Raw C0 Score: 5.29
CANCER	Raw C1 Score: 1.56	Raw C0 Score: 5.45	Raw C2 Score: 6.77
COMA	Raw C1 Score: 0.67	Raw C2 Score: 5.08	Raw C0 Score: 6.82



Conclusions

Our survival analysis demonstrates that DNR decisions are primarily driven by clinical severity and diagnostic context rather than a straightforward sociodemographic gradient. Post-MICE stratified Cox modeling reveals no significant association for sex, a minor protective effect for education, and lower DNR hazards for specific race and income categories. Sensitivity analyses via clinical-triplet and 1:1 matching confirm stark baseline socioeconomic imbalances across racial groups, yet direct DNR disparities largely weaken after rigorous clinical adjustment—though the \$25-50k income bracket consistently exhibits lower DNR odds. Ultimately, DNR pathways are shaped by an intersectional matrix of prognosis, severity, race, and socioeconomic status. Nonetheless, as an observational study, this analysis cannot prove causality or definitive discrimination, leaving a potential risk for residual confounding (Simpson's paradox). Key structural limitations include unmodeled competing risks of death within the survival framework, a fixed k=3 clustering design, and a reduced sample size with a loss of temporal resolution inherent to the matched logistic sensitivity model.